

Micro-Mouse Design Report

Please note that you are required to use the Microsoft form when preparing your report: it imposes strict formatting that must not be changed and enforces character limits for each section. **Do not disable the “Protect Form” functionality in Microsoft Word.** After completion you should convert the document to PDF for final submission.

If the form misbehaves then prepare your content in a separate application, then use copy and paste to populate it.

Student number: NKHTLO001

Brief description of design task (max. 600 characters)

A short outline of the task identified and the problem it is intended to solve.

This project was about building and testing a micromouse robot that can explore and solve a maze placed on a flat surface. During testing, we noticed that sometimes the signals from the wheel encoders were lost after small impacts or when the battery level got low. These problems caused errors in mapping and navigation. The goal was to find the causes of these signal losses, then improve the system using minor hardware fixes and software methods like signal filtering. The aim was to make the robot more reliable for autonomous operation.

Visual aids for this report

Any figures, images, or other graphics can be placed in this section and be referred to in the rest of the text. This can include excerpts of Stateflow or Simulink block diagrams. Note that you can only insert a single graphic file here, so prepare your content separately, convert to an image, and insert as required.

Below is the Simulink model with encoder data being read into the Stateflow chart for processing and a branch to make it go straight using the gyroscope.

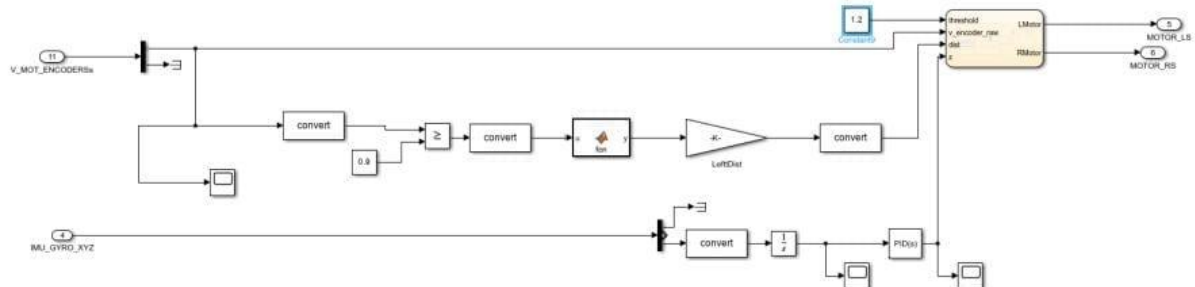


Figure 1: MATLAB Simulink Model solution

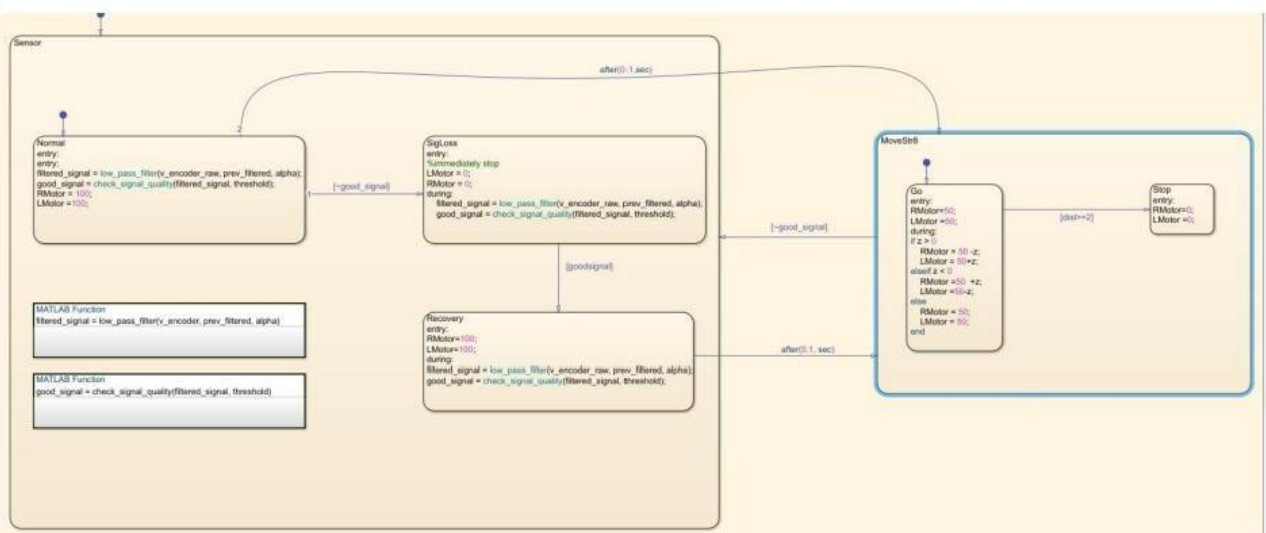


Figure 2: Stateflow chart from the Simulink Model

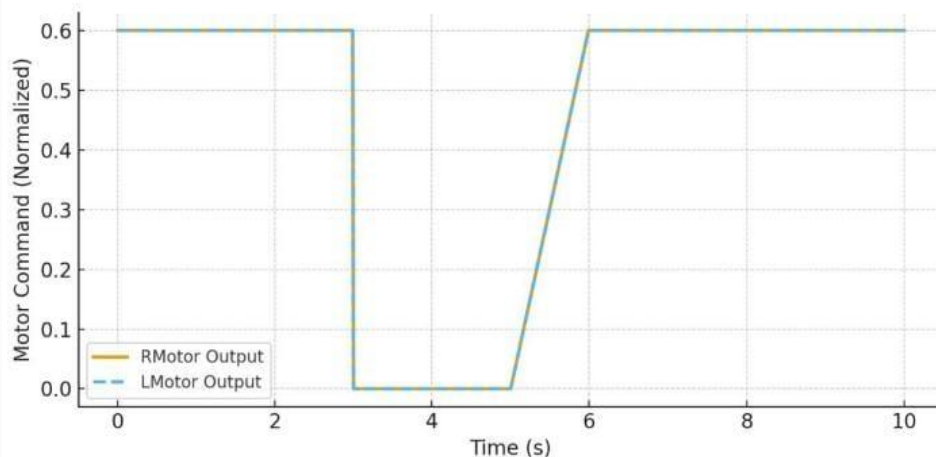


Figure 3: Motor power percentage during the different operation stages

Design criteria and constraints (max. 1000 characters)

These are fundamental elements in the engineering design process. Criteria are the desired features, functions, and performance standards that a design must meet, while constraints are the limitations or restrictions that must be considered during the design process.

The robot must accurately explore and map the maze without missing any cells. Its wheel encoder readings must remain reliable despite mild bumps and small battery voltage drops. The hardware system is constrained by the TOF sensors and gyros on the sensor board. The robot uses battery power only, with no external source during operation making it hard to monitor signals from the sensors while exploring the maze. The other constraint is to only use Simulink on Matlab to develop the software.

Main design assumptions with justification (max. 1000 characters)

Design assumptions are fundamental beliefs or statements about a system, its environment, or its users, that are taken as true without proof during the design process. Justification involves providing reasons and evidence to support these assumptions, explaining why they are reasonable and necessary for the design to proceed.

I assumed the maze and surface are level and steady, as testing is done on lab tables. The robot will occasionally hit the walls lightly during movement and turns, which can disturb the encoders. Battery voltage is expected to vary over time, especially under motor load, which can affect sensor signals. These assumptions are based on observations from test runs and standard lab environments. I also assumed that the maximum rating for the encoders was 1V as seen from the simulation and through rapid development. This assumption helps me pick a threshold voltage for when the sensor is not having normal readings.

Design process description (max. 1000 characters)

The design process is a structured approach, often iterative, used to solve problems and develop solutions. It typically involves identifying a need, researching the problem, brainstorming ideas, creating prototypes, testing, and refining the design until it meets the desired outcome.

To address unreliable encoder signals caused by noise and environmental factors, I developed a layered Stateflow control system for robust robot movement. The system integrates sensor filtering and discrete state management—Normal, Signal Loss, and Recovery—to maintain consistent motor control. MATLAB Function blocks implement a low-pass filter and signal quality check, evaluating encoder inputs each cycle to decide whether to act or transition to a safer state. Thresholds and filter coefficients were carefully tuned to balance responsiveness with stability, avoiding overreaction to transient glitches. Real-time observation via MATLAB scopes enabled iterative refinement and validation under varied test conditions.

Design implementation (max. 1000 characters)

Design implementation describes the process of translating a design into a functional reality. It's the stage where the theoretical design is brought to life, whether it's a software application, a physical product, or a system.

In the Stateflow chart, sensor logic begins in the Normal state, where encoder values are filtered and checked in real time. If readings are steady and above threshold, the robot's motors ("LMotor," "RMotor") respond directly to the filtered signals for smooth control. If noise or signal loss is detected, the system transitions to "SigLoss," halting both motors to prevent erratic behavior while continuing to monitor input quality. Upon detecting reliable signals, it enters "Recovery," where motor control resumes—potentially ramped—and signal quality is re-evaluated before returning to Normal. This layered design integrates robust filtering with logical state transitions, ensuring resilience against sensor glitches and environmental disturbances.

Design evaluation and testing (max. 1000 characters)

Evaluation and testing is crucial for ensuring a design meets its objectives and user needs. This process involves assessing a design's feasibility, desirability, and viability, and identifying areas for improvement before it gets deployed.

To assess system performance, five structured tests were conducted using a rapid Simulink model and MATLAB scopes. In each test, encoder signals were deliberately disrupted—via sensor blockage or electrical noise—while the robot was commanded to move straight. The system entered the signal-loss state within one control cycle in all cases, setting both RMotor and LMotor to zero (Figure 3, at 3 seconds). Upon signal recovery, straight-line motion resumed within one second. These results confirm that the Stateflow algorithm met the criteria for fast fault detection and recovery, operating reliably under real-world disturbances without hardware modifications.

Instructions

This report should document a structured design process that you conducted in relation to the micro-mouse project. You should explicitly be demonstrating use of methods presented to you in EEE3088F. The main purpose is to provide sufficient evidence for us to claim your proficiency in the ECSA Design Graduate Attribute (GA3), which is required to pass the course and complete your BSc(Eng) programme.

You can choose any design task for which you can confidently present evidence of design thinking. Without being prescriptive, some examples might be to design, build, and test:

1. a more realistic and operationally useful Simulink model of the micro-mouse motor drive system
2. a realistic Simulink model for time-of-flight distance measurements that function specifically in a maze environment
3. Simulink model enhancements that accommodate a realistic perturbation such as friction, wheel slip, variable robot load (mass), operation on an inclined plane, or other
4. processes for signal conditioning on measurements (e.g. improving IMU performance through calibration and noise modelling)
5. a process for smoothing robot hardware motion (e.g. prediction and filtering of actuation signals)
6. a Kalman filter-type algorithm for combining measurements and actuator signals in a structured manner
7. a coordinated turn procedure that permits 90 degree turns of the robot even with significant forward velocity
8. a method to repurpose the line-following sensors to function as rotary encoders of the wheels
9. a method for making the robot follow a predetermined trajectory
10. an overall improved micro-mouse simulator for in-the-loop algorithm development.

The task for which you provide evidence need not align with any submission milestone, and it does not have to be novel or earth shattering. However, the approach taken and the reporting thereof must follow formal design processes at the level required and must be objectively evaluated.

In essence, you should think of a micro-mouse related problem, approach it in a design-oriented manner, implement and test a proposed solution, and document the process you followed and the outcomes.

The format for this design report is very constrained. You are given limited space to present information relevant to each evaluation criterion, and each section will be evaluated for evidence it provides of your design thinking skills. A guide to the purpose of each section is as follows:

- **Brief description of design task:** A short outline of the task identified and the problem it is intended to solve.
- **Visual aids for this report:** Any figures, images, or other graphics can be placed in this section and be referred to in the rest of the text. This can include excerpts of Stateflow or Simulink block diagrams.
- **Design criteria and constraints:** These are fundamental elements in the engineering design process. Criteria are the desired features, functions, and performance standards that a design must meet, while constraints are the limitations or restrictions that must be considered during the design process.

- **Main design assumptions with justification:** Design assumptions are fundamental beliefs or statements about a system, its environment, or its users, that are taken as true without proof during the design process. Justification involves providing reasons and evidence to support these assumptions, explaining why they are reasonable and necessary for the design to proceed.
- **Design process description:** The design process is a structured approach, often iterative, used to solve problems and develop solutions. It typically involves identifying a need, researching the problem, brainstorming ideas, creating prototypes, testing, and refining the design until it meets the desired outcome.
- **Design implementation:** Design implementation describes the process of translating a design into a functional reality. It's the stage where the theoretical design is brought to life, whether it's a software application, a physical product, or a system.
- **Design evaluation and testing:** Evaluation and testing is crucial for ensuring a design meets its objectives and user needs. This process involves assessing a design's feasibility, desirability, and viability, and identifying areas for improvement before it gets deployed.

The marking rubric for these reports is as follows:

Topic (weight)	Unacceptable (0)	Marginal (1)	Acceptable (2)	Exceptional (3)
Visual aids for this report (3)	Visual aids are unclear, confusing, or difficult to understand. Fails to effectively communicate design elements due to poor visual presentation	Visual aids are somewhat effective, but unclear or confusing in parts. May lead to misinterpretation of design elements due to poor visual presentation	Visual aids are clear and effectively communicate design elements. Meets minimum requirements for visual clarity and presentation	Visual aids are highly effective, clear, and engaging. Exceeds expectations for visual communication skills, creativity, and presentation
Design criteria and constraints (3)	Fails to identify or acknowledge critical design constraints. Ignores fundamental principles that limit design	Partially identifies some, but not all, critical constraints. Assumes factors without thorough consideration of limitations	Clearly identifies and justifies relevant design constraints. Demonstrates consideration of limitations or trade-offs in design decisions	Thoroughly identifies all critical constraints. Consistently considers design principles that account for limitations and trade-offs
Main design assumptions with justification (3)	Assumptions not identified or justified. Ignores critical variables and constraints	Lacks clear identification or justification of assumptions. Assumptions not thoroughly	Assumptions clearly identified and justified. Consistently applied problem constraints with	Assumptions rigorously identified and justified through thorough analysis. Assumptions

		explored and considered	evidence to support	explicitly made explicit and transparent
Design process description (3)	Lacks clear description of design process. Fails to demonstrate understanding of design methodology	Partially describes design process but lacks detail. Fails to identify key steps or decisions in the design process	Clearly describes main stages of the design process. Demonstrates understanding of key design concepts and principles	Comprehensive description of the entire design process. Clearly explains thought processes, decisions, and trade-offs
Design implementation (3)	Does not demonstrate successful implementation of design. Fails to demonstrate functional or effective design solution	Partially implements design, but with significant flaws. Fails to demonstrate clear and consistent implementation of design	Demonstrates successful implementation of basic design elements. Shows clear and consistent execution of design principles	Seamlessly implements design elements with attention to detail. Demonstrates exceptional execution of design principles
Design evaluation and testing (3)	Ignores or fails to evaluate design thoroughly. Fails to test and validate design assumptions	Partially evaluates design, but with significant limitations. Fails to test and validate design assumptions, or tests are not comprehensive	Consistently evaluates design for functionality and usability. Tests and validates design assumptions	Thoroughly evaluates design for all aspects. Validates design assumptions through rigorous testing and analysis

A discretionary further 2 points will be awarded for overall quality of the work and coherence of the report.

Note that to satisfy the GA assessment you will be required to get at least 1.5 marks out of 3 for every topic category in the rubric (i.e. no “Unacceptable” anywhere in the report, and “Marginal” is only permitted if an additional 0.5 marks is added via the rubric). This is separate from the mark that you receive, and a passing mark does not indicate that you’ve successfully attained the relevant GA.